

Lecture 19

Bayesian Linear Models Part 1

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(Edited from the lecture notes by Brian J. Reich, NC State)



Linear regression

- ▶ Linear regression is by far the most common statistical model.
- ▶ It includes as special cases the t-test and ANOVA.
- ▶ The multiple linear regression model is

$$Y_i \sim \text{Normal}(\beta_0 + X_{i1}\beta_1 + \dots + X_{ip}\beta_p, \sigma^2),$$

independently across the $i = 1, \dots, n$ observations.

- ▶ As we'll see, Bayesian and classical linear regression are similar if $n \gg p$ and the priors are non-informative.
- ▶ However, the results can be different for challenging problems, and the interpretation is different in all cases.

Outline of Chapter 4

- ▶ Bayesian t-tests
- ▶ Bayesian linear regression
 - ▶ Gaussian priors
 - ▶ Jeffreys' priors
 - ▶ Shrinkage priors
- ▶ Generalized linear models
- ▶ Random effects
- ▶ Flexible linear models
 - ▶ Non-linear regression
 - ▶ Heteroskedastic errors
 - ▶ Non-Gaussian errors
 - ▶ Correlated errors

Bayesian one-sample (i.e., paired) t-test

- ▶ Say $Y_1, \dots, Y_n \sim \text{Normal}(\mu, \sigma^2)$.
- ▶ Typically Y_i is the difference of a pair of measurements, e.g., the post- minus pre-test for subject i .
- ▶ Therefore the interest is to compare μ to zero.
- ▶ We will consider two cases: σ^2 known and σ^2 unknown.

Jeffreys priors for the model $Y \sim \text{Normal}(\mu, 1)$

- ▶ The log likelihood is

$$\ell(\theta) = \log[f(Y|\theta)] = -\frac{1}{2} \log(2\pi) - \frac{1}{2}(Y - \mu)^2.$$

- ▶ The first derivative is

$$\ell'(\theta) = Y - \mu.$$

- ▶ The second derivative is

$$\ell''(\theta) = -1.$$

- ▶ Therefore,

$$I(\theta) = -E(\ell''(\theta)) = 1.$$

- ▶ Finally, the prior is

$$\pi(\theta) \propto I(\theta)^{1/2} = 1.$$

Bayesian one-sample (i.e., paired) t-test

- ▶ For fixed σ , the Jeffreys' prior for μ is $\pi(\mu) = 1$.
- ▶ The posterior is

$$\begin{aligned} p(\mu|\mathbf{Y}) &\propto \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \mu)^2 \right] \\ &\propto \exp \left[-\frac{n\mu^2}{2\sigma^2} \right] \times \exp \left[\frac{\mu \sum_{i=1}^n Y_i}{\sigma^2} \right] \\ &\propto \exp \left[-\frac{n}{2} (\mu - \bar{Y})^2 \right]. \end{aligned}$$

By matching the terms with those in the density of a normal distribution, we have

$$\mu|\mathbf{Y}, \sigma \sim \text{Normal} \left(\bar{Y}, \frac{\sigma^2}{n} \right).$$

Bayesian one-sample (i.e., paired) t-test

- ▶ Therefore the posterior mean is the sample mean,

$$E(\mu|\mathbf{Y}) = \bar{Y}.$$

- ▶ The 95% credible set is

$$\bar{Y} \pm 1.96 \frac{\sigma}{\sqrt{n}}.$$

- ▶ For the test of $\mathcal{H}_0 : \mu \leq 0$ versus $\mathcal{H}_1 : \mu > 0$,

$$\text{Prob}(\mathcal{H}_0|\mathbf{Y}) = \text{Prob}(\mu \leq 0|\mathbf{Y}) = \Phi(\sqrt{n}\bar{Y}/\sigma)$$

is the frequentist p -value.

Jeffreys priors for σ for the model $Y \sim \text{Normal}(0, \sigma^2)$

- ▶ The log likelihood is

$$\ell(\theta) = \log[f(Y|\theta)] = -\frac{1}{2} \log(2\pi) - \log(\sigma) - \frac{1}{2} \frac{Y^2}{\sigma^2}.$$

- ▶ The first derivative is

$$\ell'(\theta) = -\frac{1}{\sigma} + \frac{Y^2}{\sigma^3}.$$

- ▶ The second derivative is

$$\ell''(\theta) = \frac{1}{\sigma^2} - 3\frac{Y^2}{\sigma^4}.$$

- ▶ Therefore,

$$I(\theta) = -E(\ell''(\theta)) = -\frac{1}{\sigma^2} + 3\frac{\sigma^2}{\sigma^4} = \frac{2}{\sigma^2}.$$

- ▶ Finally, the prior is

$$\pi(\theta) \propto I(\theta)^{1/2} \propto \frac{1}{\sigma}.$$

Jeffreys priors for σ^2 for the model $Y \sim \text{Normal}(0, \sigma^2)$

- ▶ The log likelihood is

$$\ell(\theta) = \log[f(Y|\theta)] = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log(\sigma^2) - \frac{1}{2} \frac{Y^2}{\sigma^2}.$$

- ▶ The first derivative is

$$\ell'(\theta) = -\frac{1}{2} \frac{1}{\sigma^2} + \frac{1}{2} \frac{Y^2}{(\sigma^2)^2}.$$

- ▶ The second derivative is

$$\ell''(\theta) = \frac{1}{2} \frac{1}{(\sigma^2)^2} - \frac{Y^2}{(\sigma^2)^3}.$$

- ▶ Therefore,

$$I(\theta) = -E(\ell''(\theta)) = -\frac{1}{2} \frac{1}{\sigma^4} + \frac{\sigma^2}{\sigma^6} = \frac{1}{2} \frac{1}{\sigma^4}.$$

- ▶ Finally, the prior is

$$\pi(\theta) \propto I(\theta)^{1/2} \propto \frac{1}{\sigma^2}.$$

Jeffreys priors for $\theta = (\mu, \sigma^2)'$ for the model $Y \sim \text{Normal}(\mu, \sigma^2)$

- ▶ The log likelihood is

$$\ell(\theta) = \log[f(Y|\theta)] = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log(\sigma^2) - \frac{1}{2} \frac{(Y - \mu)^2}{\sigma^2}.$$

- ▶ The first derivative is

$$\ell'_1(\theta) = \frac{Y - \mu}{\sigma^2}, \quad \ell'_2(\theta) = -\frac{1}{2} \frac{1}{\sigma^2} + \frac{1}{2} \frac{(Y - \mu)^2}{(\sigma^2)^2}$$

- ▶ The second derivative is

$$\ell''_{11}(\theta) = -\frac{1}{\sigma^2}, \quad \ell''_{12}(\theta) = \ell''_{21}(\theta) = -\frac{Y - \mu}{(\sigma^2)^2}, \quad \ell''_{22}(\theta) = \frac{1}{2} \frac{1}{(\sigma^2)^2} - \frac{(Y - \mu)^2}{(\sigma^2)^3}.$$

Jeffreys priors for $\theta = (\mu, \sigma^2)'$ for the model $Y \sim \text{Normal}(\mu, \sigma^2)$

► Therefore,

$$-E(\ell''_{11}(\theta)) = \frac{1}{\sigma^2}$$

$$-E(\ell''_{12}(\theta)) = -E(\ell''_{21}(\theta)) = 0$$

$$-E(\ell''_{22}(\theta)) = \frac{1}{2} \frac{1}{\sigma^4}.$$

► Finally, the prior is

$$\pi(\theta) \propto |I(\theta)|^{1/2} \propto \frac{1}{\sigma^3}.$$

Bayesian one-sample (i.e., paired) t-test

- ▶ When σ^2 is unknown, the Jeffreys' prior is

$$\pi(\mu, \sigma^2) \propto \left(\frac{1}{\sigma^2} \right)^{3/2}.$$

- ▶ The marginal posterior integrating over uncertainty in σ^2 is

$$\mu | \mathbf{Y} \sim t_n \left(\bar{Y}, \frac{\hat{\sigma}^2}{n} \right),$$

where $\hat{\sigma}^2 = \sum_{i=1}^n (Y_i - \bar{Y})^2 / n$.

- ▶ This is very similar to the frequentist t -test, except that the degrees of freedom is n rather than $n - 1$.
- ▶ This is the effect of the prior.

Bayesian two-sample t-test

- Say the n_1 observations from group 1 are

$$Y_i \sim \text{Normal}(\mu, \sigma^2),$$

and the n_2 observations from group 2 are

$$Y_i \sim \text{Normal}(\mu + \delta, \sigma^2).$$

- The goal is to compare δ to zero.
- With σ^2 known and Jeffrey's prior $\pi(\mu, \delta) = 1$,

$$\delta | \mathbf{Y}, \sigma^2 \sim \text{Normal} \left(\bar{Y}_2 - \bar{Y}_1, \frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2} \right),$$

and the results are identical to the two-sample z-test.

Bayesian two-sample t-test

- ▶ When σ^2 is unknown, the Jeffreys' prior is

$$\pi(\mu, \delta, \sigma^2) \propto \left(\frac{1}{\sigma^2}\right)^2.$$

- ▶ The marginal posterior integrating over uncertainty in σ^2 and μ is

$$\delta|\mathbf{Y} \sim t_n \left(\bar{Y}_2 - \bar{Y}_1, \frac{\hat{\sigma}^2}{n_1} + \frac{\hat{\sigma}^2}{n_2} \right),$$

where the pooled variance estimator is

$$\hat{\sigma}^2 = \left[\sum_{i=1}^{n_1} (Y_i - \bar{Y}_1)^2 + \sum_{i=n_1+1}^{n_2} (Y_i - \bar{Y}_2)^2 \right] / n.$$

- ▶ This is very similar to the frequentist t -test, except that the degrees of freedom is $n = n_1 + n_2$ rather than $n - 2$.
- ▶ This is the effect of the prior.

Review of least squares

- ▶ The least squares estimate of $\beta = (\beta_0, \beta_1, \dots, \beta_p)^T$ is

$$\hat{\beta}_{OLS} = \underset{\beta}{\operatorname{argmin}} \sum_{i=1}^n (Y_i - \mu_i)^2,$$

where $\mu_i = \beta_0 + X_{i1}\beta_1 + \dots + X_{ip}\beta_p$.

- ▶ $\hat{\beta}_{OLS}$ is unbiased even if the errors are non-Gaussian.
- ▶ If the errors are Gaussian, then the likelihood is proportional to

$$\prod_{i=1}^n \exp \left[-\frac{(Y_i - \mu_i)^2}{2\sigma^2} \right] = \exp \left[-\frac{\sum_{i=1}^n (Y_i - \mu_i)^2}{2\sigma^2} \right].$$

- ▶ Therefore, if the errors are Gaussian $\hat{\beta}_{OLS}$ is also the MLE.

Review of least squares

- ▶ Linear regression is often simpler to describe using linear algebra notation.
- ▶ Let $\mathbf{Y} = (Y_1, \dots, Y_n)^T$ be the response vector and $\mathbf{X}$ be the $n \times (p + 1)$ matrix of covariates.
- ▶ Then the mean of $\mathbf{Y}$ is $\mathbf{X}\beta$ and the least squares solution is

$$\hat{\beta}_{OLS} = \underset{\beta}{\operatorname{argmin}} (\mathbf{Y} - \mathbf{X}\beta)^T (\mathbf{Y} - \mathbf{X}\beta) = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}.$$

- ▶ If the errors are Gaussian then the sampling distribution is

$$\hat{\beta}_{OLS} \sim \text{Normal} \left[\beta, \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1} \right].$$

- ▶ If the variance σ^2 is estimated using the mean squared residual error, then the sampling distribution is multivariate t .

Bayesian regression

- ▶ The likelihood remains

$$Y_i \sim \text{Normal}(\beta_0 + X_{i1}\beta_1 + \dots + X_{ip}\beta_p, \sigma^2)$$

independent for $i = 1, \dots, n$ observations

- ▶ As with a least squares analysis, it is crucial to verify this is appropriate using qq-plots, added variable plots, etc.
- ▶ A Bayesian analysis also requires priors for β and σ
- ▶ We will focus on prior specification since this piece is uniquely Bayesian.

Thank you!